MILITARY SPECIFICATION

STEEL BAR, AND WIRE, (455) CORROSION RESISTANT, PRECIPITATION HARDENING

1. SCOPE

1.1 Scope. This specification covers one composition of corrosion-resistant, precipitation hardening stainless steel in bar, wire, and forging stock.

1.2 Classification.

1.2.1 Solution heat treated condition. Unless otherwise specified, the material shall be furnished in the solution heat treated condition.

1.2.2 Surface condition. The material shall be furnished in one of the following conditions, as specified (see 6.2):

(1) Scale not removed (hot finished)

(2) Pickled, blast cleaned or otherwise descaled

(3) Cold drawn

(4) Turned

(5) Centerless ground

(6) Polished

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

FSC 9510

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STANDARDS

Federal

Federal Test Method Standard No. 151	Metals; Test Methods
FED-STD-183	Continuous Identification Marking of Iron and Steel Products

Military

MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

Society of Automotive Engineers, Incorporated (AMS)

2241	Tolerances, Corrosion and Heat Resistant Steel Bars and Steel Bars and Wire, and Titanium and Titanium Alloy Bars and Wire

(Copies of Aerospace Materials Specifications should be obtained from the Society of Automotive Engineers, Inc., 485 Lexington Avenue, New York, New York 10017.)

American Society for Testing and Materials (ASTM)

E 8	Methods of Tension Testing of Metallic Materials
E 18	Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Copies of ASTM standards should be obtained from the Society of Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference for libraries. They are also distributed among technical groups and using Federal agencies.

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3. REQUIREMENTS

3.1 Chemical Composition. The chemical composition shall be as specified in table I.

TABLE I. Chemical composition

Element	Percent by Weight	Check analysis tolerance 1/ Under min.	Over max.
Carbon	0.05 (max)	--	0.01
Manganese	0.50 (max)	--	.03
Phosphorus	0.025 (max)	--	.005
Sulfur	0.025 (max)	--	.005
Chromium	11.00-12.50	0.15	.15
Nickel	7.50-9.50	.10	.10
Silicon	0.50 (max)	--	.15
Columbium + Tantalum	0.10-0.50	.05	.05
Titanium	0.80-1.40	.05	.05
Copper	1.50-2.50	.07	.07
Molybdenum	0.50 (max)	--	.03
Iron	Balance	--	--

1/ The average of all the separate determinations shall be within the limits specified in the "Percent by Weight" column. Individual determinations may vary to the extent shown in the "Check analysis tolerance" column, except that the several determinations of a single element in any one heat shall not vary both above and below the specified range.

3.2 Solution heat treatment. Solution heat treating shall be accomplished by heating to 1525° ±25°F for 30 minutes and quenching in water.

3.3 Mechanical properties. The properties of solution treated material shall conform to 3.3.1, 3.3.2 and 3.3.3.

3.3.1 Bars: Shall have hardness not higher than Brinell 331 or equivalent when taken midway between surface and center.

3.3.2 Wire: Shall have tensile strength not higher than 175,000 psi.

3.4 Response to precipitation heat treatment. Solution treated material shall, after the heat treatment outlined below, meet the requirements for condition H 950 material, as shown in table II. Heat treatment shall be as follows: Heat to 950°F ±10°F for 4 hours and air cool.

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TABLE II. Mechanical properties, H 950 condition

Tensile Strength (psi)	Yield strength psi, 0.2 percent offset. 1/	Elongation percent of gage length.	Reduction area percent	Brinnel hardness number
220,000 (min)	205,000 (min)	10 (min)	40 (min)	415 (min)

1/ Or measured at 0.0184 inch per inch extension under load.

3.5 Dimensional tolerances. The variation from nominal dimensions shall not exceed the tolerance limits specified by AMS 2241. Tolerances for sizes and forms not covered by AMS 2241 shall be agreed upon by purchaser and vendor.

3.6 Identification of product. Unless otherwise specified, material shall be marked for identification in accordance with Federal Standard No. 183. The legend shall include the following items.

MIL-S- 83311(USAF)
Manufacturer's name or trademark, (optional).
455 Steel
Condition
Heat number.

3.7 Workmanship. Material shall be sound, of uniform quality and condition, free from pipes, laps, cracks, twists, seams, or other defects detrimental to the performance of parts fabricated from the material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary

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to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. The examination and testing of the steel shall be classified as quality conformance inspection.

4.3 Examination of product. Each piece of bar and wire shall be visually examined for compliance with the surface conditions and workmanship requirements.

4.3.1 Samples selected in accordance with table III shall be examined for compliance with nominal dimensions, identification marking, and preparation for delivery.

Table III. Sample size for examination

Lot size	Sample size	Sampling plan acceptance number
1 to 25	4	0
26 to 65	6	0
66 to 180	8	0
181 to 300	9	0
301 to 800	10	0
Over 800	15	0

4.4 Chemical analysis.

4.4.1 Sampling. Samples for chemical analysis shall be selected in accordance with Method 111 of Federal Test Method Standard No. 151 to represent each heat in the shipment. The sample shall consist of not less than 2 ounces of material.

4.4.2 Samples for chemical analysis may be waived provided that all of the material under inspection can be identified as being made from a heat previously analyzed and found to be in accordance with the chemical composition specified herein.

4.4.3 Method. Chemical analysis shall be made in accordance with standard wet chemical, spectrochemical, or other analytical methods. In the event of dispute, analysis shall be by wet chemical methods.

4.5 Mechanical Properties.

4.5.1 Sampling. One or more test samples shall be selected from each lot submitted for acceptance. A lot is defined as material of the same

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heat, of the same condition, of the same thickness, and submitted for acceptance at one time.

4.5.2 Tensile Test Method. Tensile tests shall be conducted in accordance with applicable requirements of ASTM E8.

4.5.3 Hardness Test Method. Hardness tests hall be conducted in accordance with applicable requirements of ASTM E 18.

4.6. Response to precipitation heat treatment.

4.6.1 Sampling. One or more samples shall be selected from each heat submitted for acceptance.

4.6.2 Tensile Test Method. Tensile tests shall be conducted in accordance with applicable requirements of ASTM E 8.

4.6.3 Hardness Test Method. Hardness tests shall be conducted in accordance with applicable requirements of ASTM E 18.

4.7 Rejection. When the failure of one or more specimens indicate that the test sample fails to meet the specified requirement, the entire lot shall be rejected.

4.7.1 Retest. At the discretion of the supplier, retest will be permitted. A retest sample of five specimens shall be tested to replace each failed specimen of the initial sample. If one or more of the retest specimens fail, the lot shall be rejected with no further retesting permitted.

4.7.2 Resubmittal. If the defectives in a lot previously rejected can be identified and removed from the lot, resampling at the frequency of the initial sampling will be acceptable.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. The steel shall be preserved and packaged in accordance with MIL-STD-163.

5.1.2 Level C. The steel shall be preserved and packaged in accordance with standard commercial practice.

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5.2 Packing.

5.2.1 Level A. The steel shall be packed in accordance with MIL-STD-163.

5.2.2 Level C. The steel shall be prepared for shipment in accordance with commercial practice to ensure carrier acceptance and safe transportation at the lowest rate to the point of delivery and shall meet, as a minimum, the requirements of carrier rules and regulations applicable to the mode of transportation.

5.3 Marking of shipments. Each shipping container shall be marked with information listed below, and marked to indicate method of packing. Shipping containers shall be marked in accordance with MIL-STD-163.

STEEL BAR, AND WIRE, (455) CORROSION RESISTANT
PRECIPITATION HARDENING.

MIL-S- 83311(USAF)
Size, and surface finish
Heat number
Quantity contained (as defined in the contract or order for each size).
Name of manufacture
Name of contractor (if different from manufacturer).
Contract or Order Number.

6. NOTES

6.1 Intended use. The steel is intended for use in applications which require moderately high corrosion resistance and mechanical properties.

6.1.1 Suggested fabrication techniques. Parts may be readily fabricated in the solution heat treated (as received) condition, and subsequently heat treated in accordance with 3.4 to obtain the properties of table II.

6.2 Ordering data. Procurement documents should specify:

(1) Title, number, and date of this specification.
(2) Size
(3) Surface condition (see 1.2.2)
(4) Levels of packaging and packing (see 5.1 and 5.2).

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Project No. 9510-F002

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

AFLC–WPAFB–OCT 67 2M

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